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ExonViz: Transcript visualization made easy

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DOI: 10.xxxxxx/draft

Software

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Editor:

Submitted: 17 October 2025
Published: unpublished

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Summary

Transcripts of a gene contain one or more **exons**, which encode the functional parts of the transcript, and **introns**, which are removed in a process called **splicing**. A single gene typically encodes multiple **transcripts** by including different exons. Protein coding genes include one or more coding exons, which encode the protein using three-letter sequences called **codons**. It is important to realize that exon boundaries and codon boundaries do not coincide, *i.e.* a codon can span two exons. If the codon boundaries of adjacent exons are not aligned, an often detrimental **frame shift** is introduced. Taking exon boundary reading frames, which we will call **exon boundary frames** into account is crucial when considering the effect of mutations and when designing genetic therapies.

`ExonViz` is a Python package and web application that creates biologically accurate RNA transcript figures, including features such as coding regions, genetic variants and exon boundary frames. The ExonViz website has been used to generate over 8000 transcript figures between September 2023 and September 2025.

ExonViz is written in Python 3, its web interface is build using Flask. It uses the public Mutalyzer API (Lefter et al., 2021) to fetch transcript annotations. This gives ExonViz access to all transcripts defined in the RefSeq (O'Leary et al., 2016) and Ensembl (Harrison et al., 2023) databases across many species, ranging from human and mouse to fruit fly and coelacanth. Transcripts and annotations defined on the reverse strand are reversed, so ExonViz always visualizes transcripts in their forward orientation.

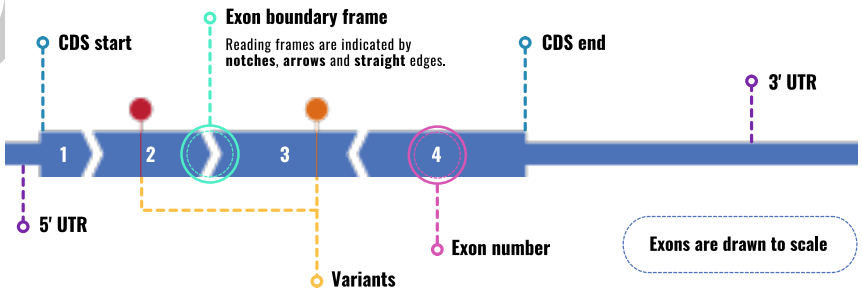

Figure 1: Example transcript highlighting ExonViz features. **5' UTR**: Non coding region at the start of the transcript. **CDS start**: Start of the coding region. **CDS end**: End of the coding region. **3' UTR**: Non coding region at the end of the transcript.

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Statement of need

Visualization of transcripts, including features like exon boundary frames, coding and non coding regions is important within the field of clinical and human genetics (Walker et al., 2023). Illustrating the exon structure and the location of variants is common practice, especially when new genes or transcripts have been discovered. These illustrations are also used to assess potential genetic treatment options (e.g., canonical exon skipping), in teaching settings, in diagnostics, to identify mutational hotspots and for genetic counseling. To date, most people have to resort to manually drawing transcripts with tools like Illustrator, Photoshop or BioRender, or forgo illustrations altogether. Some tools have been made available that aid in drawing transcripts (ggtranscript (Gustavsson et al., 2022) and wiggleplotr (Alasoo, 2017)), visualize different transcript isoforms (genepainter (Mühlhausen et al., 2015)), or visualize variants (Variant View (Ferstay et al., 2013)). However, none of these tools can draw exon boundary frames. Knowledge about the exon boundary frames aids in the assessment of the pathogenicity of genetic variants using the ACMG-AMP guidelines (Richards et al., 2015) when evaluating exon spanning deletions (Cheerie et al., 2025) and when interpreting the effects of splice altering variants (Walker et al., 2023). Creating transcript visualizations must be quick and easy if they are to be utilized in clinical and day to day settings, rather than to create a bespoke figure for a manuscript or presentation.

To our knowledge, there currently are no easily usable tools which allow the user to draw all features required for a comprehensive overview of a transcript's structure and the location of variants of interest.

Method

ExonViz visualizes the exon boundary frames by using different shapes for the boundary of exons. It is important that adjacent coding exons have aligning codon boundaries, to prevent a shift in the reading frame of the protein. Figure 2 shows all possible combinations of exon and codon boundaries, and the corresponding exon boundary shapes. When the exon and codon boundaries coincide (frame 0) the exons are drawn with a straight edge, as is the case of exon 1 and 2. Exon 2 ends one base into the codon (in frame 1), which is drawn using an arrow on the end of the exon. Exon 3 starts in frame 1, and is drawn with a notch at the start of the exon. This reversed for the boundary between exons 3 and 4, which is in frame 2. Since the exons of a transcript should fit together, exons in conflicting frames (*e.g.* because of a frame shift inducing variant) are easily spotted due to the fact that the exon boundaries do not fit together.

Figure 2: Visualization of the relation between codons and exon boundary frames. The shapes of the exons illustrate the relation between the exon boundaries and the codon boundaries.

The output of ExonViz is an SVG figure generated using the svg-py library, which can be used directly or modified using modern graphical editing programs. It is also possible to output the transcript and variants in TSV format, edit the transcript using any text editor or spreadsheet program, and draw the modified transcript using ExonViz. The online documentation has a number of examples of custom transcripts that can be visualized this way.

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Conclusion

To our knowledge, ExonViz is the first publicly available application that allows for automatic visualization of transcripts with additional features such as exon boundary frames and variants along the transcript. ExonViz can be used for illustrations within publications, assessment of treatment options, teaching purposes and genetic counseling. Figures generated by ExonViz are free to use under the Creative Commons BY license . Furthermore, we allow the user to construct their own transcripts, for example to visualize non-standard exons or alternative isoforms. ExonViz can be accessed as a web application via exonviz.rnatherapy.nl or installed via PyPI. The source code is available on Github.

Acknowledgments

We would like to thank the members of the Dutch Center for RNA Therapeutics for their ideas, suggestions and their feedback on earlier versions of ExonViz. We also thank Maximilian Haeussler and his colleagues at the UCSC for their efforts implementing exon boundary frame information into the UCSC Genome Browser. We also thank Nanieke van den Berg for creating Figure 1.

References

Alasoo, K. (2017). Wiggleplotr: Make read coverage plots from BigWig files. *Bioconductor. Doi*, *10*, B9.

Cheerie, D., Meserve, M. M., Beijer, D., Kaiwar, C., Newton, L., Tavares, A. L. T., Verran, A. S., Sherrill, E., Leonard, S., Sanders, S. J., & others. (2025). Consensus guidelines for assessing eligibility of pathogenic DNA variants for antisense oligonucleotide treatments. *The American Journal of Human Genetics*. https://doi.org/10.1016/j.ajhg.2025.02.017

Ferstay, J. A., Nielsen, C. B., & Munzner, T. (2013). Variant view: Visualizing sequence variants in their gene context. *IEEE Transactions on Visualization and Computer Graphics*, *19*(12), 2546–2555. https://doi.org/10.1109/TVCG.2013.214

Gustavsson, E. K., Zhang, D., Reynolds, R. H., Garcia-Ruiz, S., & Ryten, M. (2022). Ggtranscript: An r package for the visualization and interpretation of transcript isoforms using ggplot2. *Bioinformatics*, *38*(15), 3844–3846. https://doi.org/10.1093/bioinformatics/btac409

Harrison, P. W., Amode, M. R., Austine-Orimoloye, O., Azov, A. G., Barba, M., Barnes, I., Becker, A., Bennett, R., Berry, A., Bhai, J., Bhurji, S. K., Boddu, S., BrancoÂ Lins, P. R., Brooks, L., Ramaraju, S. B., Campbell, L. I., Martinez, M. C., Charkhchi, M., Chougule, K., … Yates, A. D. (2023). Ensembl 2024. *Nucleic Acids Research*, *52*(D1), D891–D899. https://doi.org/10.1093/nar/gkad1049

Lefter, M., Vis, J. K., Vermaat, M., Dunnen, J. T. den, Taschner, P. E. M., & Laros, J. F. J. (2021). Mutalyzer 2: next generation HGVS nomenclature checker. *Bioinformatics*, *37*(18), 2811–2817. https://doi.org/10.1093/bioinformatics/btab051

Mühlhausen, S., Hellkamp, M., & Kollmar, M. (2015). GenePainter v. 2.0 resolves the taxonomic distribution of intron positions. *Bioinformatics*, *31*(8), 1302–1304. https://doi.org/10.1093/bioinformatics/btu798

O'Leary, N. A., Wright, M. W., Brister, J. R., Ciufo, S., Haddad, D., McVeigh, R., Rajput, B., Robbertse, B., Smith-White, B., Ako-Adjei, D., & others. (2016). Reference sequence (RefSeq) database at NCBI: Current status, taxonomic expansion, and functional annotation. *Nucleic Acids Research*, *44*(D1), D733–D745. https://doi.org/10.1093/nar/gkv1189

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Richards, S., Aziz, N., Bale, S., Bick, D., Das, S., Gastier-Foster, J., Grody, W. W., Hegde, M., Lyon, E., Spector, E., & others. (2015). Standards and guidelines for the interpretation of sequence variants: A joint consensus recommendation of the american college of medical genetics and genomics and the association for molecular pathology. *Genetics in Medicine*, *17*(5), 405–423. https://doi.org/10.1038/s41525-022-00329-6

Walker, L. C., Hoya, M. de la, Wiggins, G. A., Lindy, A., Vincent, L. M., Parsons, M. T., Canson, D. M., Bis-Brewer, D., Cass, A., Tchourbanov, A., & others. (2023). Using the ACMG/AMP framework to capture evidence related to predicted and observed impact on splicing: Recommendations from the ClinGen SVI splicing subgroup. *The American Journal of Human Genetics*, *110*(7), 1046–1067. https://doi.org/10.1016/j.ajhg.2023.06.002

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